A counterexample to the HOMFLYPT–Kauffman span inequality

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November 2026

Abstract

The knot $12n_{296}$ has HOMFLYPT a-span 8 and Kauffman a-span 7. It therefore disproves the proposed universal comparison $\operatorname{span}_a P_K \le \operatorname{span}_a F_K$. We give an explicit five-strand braid, the complete polynomial coefficients, and an exact skein verification that uses only integer arithmetic. The polynomial values are already tabulated in KnotInfo; the comparison is checked directly from a diagram. Connected sums of this example give arbitrarily large differences $\operatorname{span}_a P_K - \operatorname{span}_a F_K$.

1 Conventions and statement

For a nonzero Laurent polynomial $Q(a,z)$, let

$$\operatorname{span}_a Q = \max\{i : [a^i]Q \neq 0\} - \min\{i : [a^i]Q \neq 0\},$$

where coefficients are Laurent polynomials in z. We use the HOMFLYPT normalisation

$$P_U = 1, \qquad aP_{L_+} - a^{-1}P_{L_-} = zP_{L_0}. \tag{1}$$

For computational convenience, set $H_L(v,z) = P_L(v^{-1},z)$; then $v^{-1}H_{L_+} - vH_{L_-} = zH_{L_0}$, the KnotInfo convention. Inversion of the first variable preserves its span.

The Kauffman polynomial is $F_L(a,z) = a^{-w(D)}\Lambda_D(a,z)$, where $w(D)$ is the writhe of an oriented diagram. Its regular-isotopy version satisfies

$$\Lambda_U = 1, \qquad \Lambda_{D_+} + \Lambda_{D_-} = z(\Lambda_{D_0} + \Lambda_{D_\infty}), \tag{2}$$

and a positive or negative curl multiplies Λ by a or a^{-1}, respectively. Thus a disjoint trivial circle multiplies Λ by $\delta_F = (a + a^{-1})/z - 1$. These are the original two-variable Kauffman conventions used by KnotInfo [4, 5].

Candidate (c7) in version 3 of Jabłonowski's paper [1] is $\operatorname{span}_a P_K \le \operatorname{span}_a F_K$. Appendix entries 32 and 33 define both quantities as the full difference of the extreme exponents. Version 4 proves the alternating case in Proposition 3.14; version 5 omits this candidate and assigns the label (c7) to another inequality [2, 3]. The statement considered here is the explicit full-span comparison above.

Theorem 1. *Let K be the closure of the braid in B_5*

$$\beta = \sigma_1^{-1}\sigma_2\sigma_3\sigma_4^{-1}\sigma_1^{-1}\sigma_3\sigma_4^{-1}\sigma_2^3\sigma_3\sigma_2. \tag{3}$$

Then K is a knot and

$$\operatorname{span}_a P_K = 8 > 7 = \operatorname{span}_a F_K.$$

In KnotInfo's notation, this braid represents $12n_{296}$.

2 Polynomial calculation and proof

Following the strands of (3) from top to bottom gives the closure cycle (1 2 4 5 3). There is one component, and the writhe of this diagram is 4.

Write $H_K(v,z) = \sum_j p_j(v) z^j$. The nonzero coefficients are

$$\begin{aligned} p_0(v) &= 3 - 5v^2 + 4v^4 - v^6, \\ p_2(v) &= 4 - 13v^2 + 12v^4 - 5v^6 + v^8, \\ p_4(v) &= 1 - 10v^2 + 12v^4 - 5v^6, \\ p_6(v) &= -2v^2 + 6v^4 - v^6, \\ p_8(v) &= v^4. \end{aligned}$$

For the Kauffman polynomial, put $b = a^{-1}$ and write $F_K(a,z) = \sum_j f_j(b) z^j$. Its coefficients are

j	$f_j(b)$
0	$b^6 + 4b^4 + 5b^2 + 3$
1	$-b^7 - 2b^5 + b$
2	$-6b^6 - 22b^4 - 26b^2 - 10$
3	$3b^7 + b^5 - 12b^3 - 10b$
4	$16b^6 + 41b^4 + 37b^2 + 12$
5	$-4b^7 + 11b^5 + 33b^3 + 18b$
6	$-14b^6 - 24b^4 - 16b^2 - 6$
7	$b^7 - 13b^5 - 25b^3 - 11b$
8	$3b^6 + b^4 - b^2 + 1$
9	$3b^5 + 5b^3 + 2b$
10	$b^4 + b^2$

These are the polynomial values tabulated in KnotInfo [4]; the independent computation described below reproduces every coefficient.

Proof of Theorem 1. The coefficients of v^0 and v^8 in H_K are $3 + 4z^2 + z^4$ and z^2, respectively. Both are nonzero, and the displayed polynomial has no powers of v outside $[0,8]$. Consequently $P_K(a,z) = H_K(a^{-1},z)$ has extreme exponents -8 and 0.

Every exponent of a in F_K lies in $[-7,0]$. The coefficients at the two endpoints are

$$[a^{-7}]F_K = -z + 3z^3 - 4z^5 + z^7, \qquad [a^0]F_K = 3 - 10z^2 + 12z^4 - 6z^6 + z^8.$$

Both are nonzero. The two spans are therefore 8 and 7. □

Corollary 2. *For every integer $m \geq 1$ there is a knot K_m such that*

$$\operatorname{span}_a P_{K_m} - \operatorname{span}_a F_{K_m} = m.$$

Proof. Take the connected sum of m copies of K. The normalised HOMFLYPT and Kauffman polynomials are multiplicative under connected sum. This follows from their skein relations by induction in one summand, with the unknot as the initial case. The coefficient ring $\mathbb{Z}[z,z^{-1}]$ is an integral domain, so the lowest and highest a-exponents add under multiplication. Thus the two spans of K_m are $8m$ and $7m$. □

3 Exact diagram verification

The supplement contains a complete verifier using Python's standard library. It constructs the oriented encounter lists of the closed braid and evaluates (1) and (2) by skein recursion. An encounter records its crossing label, whether it is over or under, and the crossing sign.

Fix an ordering and a starting point for each component. Traverse the components in that order. A crossing is bad if its first encounter is under. If there is a bad crossing, choose the first one. Switching it reduces the number of bad crossings by one, without changing the number of crossings. Either smoothing reduces the number of crossings. Hence recursion terminates by lexicographic induction on the number of crossings and then the number of bad crossings.

A diagram with no bad crossings is a descending unlink. Standard descending-diagram reduction removes it by Reidemeister moves, leaving its self-curls accounted for by the writhe. For r components the initial values are therefore

$$H = \delta_H^{r-1}, \quad \delta_H = (v^{-1} - v)/z, \qquad \Lambda = a^w \delta_F^{r-1}.$$

At a crossing of sign $h \in \{1, -1\}$, let D' be its switched diagram and D_o its oriented smoothing. The HOMFLYPT recursion is

$$H_D = v^{2h} H_{D'} + h v^h z H_{D_o}.$$

The Kauffman recursion is

$$\Lambda_D = z(\Lambda_{D_o} + \Lambda_{D_u}) - \Lambda_{D'},$$

where D_u is the other smoothing. Finally multiply by a^{-w}.

The independent implementation represents each encounter by an incoming and an outgoing port. Diagram arcs join successive encounters. At the two encounters of a removed crossing, the oriented smoothing pairs each incoming port with the other outgoing port. The other smoothing pairs the two incoming ports and the two outgoing ports. All ports then have degree two, so tracing cycles reconstructs the components. If one branch of a surviving crossing is traversed in the reverse direction, its sign is reversed; if both branches are reversed, its sign is unchanged. This supplies the crossing signs needed for every descendant.

Polynomials are sparse dictionaries of integer coefficients indexed by the two exponents. No rounding or numerical root computation occurs. Starting from (3), the computation visits 1,973 distinct HOMFLYPT states and 104,163 Kauffman states and yields the displayed polynomials. A separate signed-Gauss-code implementation [6], using the published planar-diagram notation, gives the same values.

The verifier also checks the trefoil and figure-eight formulas, the braid relation, a cancelling generator pair, cyclic conjugation, and both signs of Markov stabilisation. Run it from the supplement by

```
python src/verify_all.py
```

It reconstructs the polynomials, compares every coefficient with the included source data, and checks the two extreme exponents. Its output records the verification result and hashes of the source and data files.

4 Attribution and scope

The knot and both polynomial values belong to the existing KnotInfo tables. This note gives their direct consequence for the stated full-span inequality and an independently implemented verification from an explicit braid. One counterexample settles the universal question; a census assumption is unnecessary. The alternating-knot result of [2] is consistent with this nonalternating example.

References

[1] M. Jabłonowski, *Integer Knot Invariants: Inequalities, Computations, and Open Problems*, arXiv:2605.22652v3, Section 3, (c7), and Appendix entries 32–33. `https://arxiv.org/html/2605.22652v3`.

[2] M. Jabłonowski, same title, version 4, 12 August 2026, Proposition 3.14. `https://arxiv.org/html/2605.22652v4`.

[3] M. Jabłonowski, same title, version 5, 15 September 2026, Section 5. `https://arxiv.org/html/2605.22652v5`.

[4] C. Livingston and A. H. Moore, *KnotInfo: Table of Knot Invariants*, entry $12n_{296}$, polynomial workbook, accessed 4 November 2026. `https://knotinfo.org/homelinks/database_download.php`.

[5] KnotInfo, *Combinatorial definitions of polynomial invariants*. `https://knotinfo.org/descriptions/jones_homfly_kauffman_description/polynomial_defn.html`.

[6] A. De Lucreziis, *Implementation of the Kauffman polynomial in Python*, software repository `aziis98/kauffman-polynomial`, commit `ded3fd2d56fe4953ec42fffe659bf3e6115033d0`. `https://github.com/aziis98/kauffman-polynomial`.